Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 25

Counterexamples

Covers Chartrand & Zhang, Section 3.5

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Check the hypothesis first

A counterexample to $P(x) \Rightarrow Q(x)$ needs $P(a)$ **true** and $Q(a)$ **false**. Verify the hypothesis *before* the conclusion — an element failing the hypothesis proves nothing.

Where counterexamples hide

Zero. Negative numbers. Fractions, when the domain is $\mathbb{R}$. The empty set. Equal elements, when a claim assumes distinct ones. Try these before anything else.

Three mistakes to expect

1. **Offering an element that fails the hypothesis.**
2. **Describing a counterexample instead of naming one.** "Some negative number" is not an element.
3. **Disproving a biconditional without saying which direction failed.**

Part II: Practice Problems

Section A — Disprove

1. Disprove each by giving a counterexample, and verify both halves.

(a) Let $n \in \mathbb{Z}$. If n^2 is divisible by 4, then n is divisible by 4.

(b) Let $x \in \mathbb{R}$. If $x^2 = 9$, then $x = 3$.

(c) Let $n \in \mathbb{Z}$. If n is prime, then $n + 2$ is prime.

2. For each attempted disproof, decide whether it is valid. If not, say why and supply a valid counterexample.

(a) Claim: *if n is odd, then n^2 is odd.* Offered: $n = 2$, since 4 is even.

(b) Claim: *if $n > 0$, then $n^2 > n$.* Offered: $n = 1$.

Section B — Biconditionals and sets

3. Disprove: let $m, n \in \mathbb{Z}$; then $m + n$ is odd if and only if m and n are both odd. Say which direction fails.

4. Disprove: for all sets A, B, C, $\quad A \cup (B \cap C) = (A \cup B) \cap C$.

Section C — Prove or disprove

5. Decide whether each is true or false. Prove it or give a counterexample.

(a) Let $n \in \mathbb{Z}$. If n^2 is even, then n is even.

(b) Let $x \in \mathbb{R}$. If $x > 1$, then $x^2 > x$.

(c) Let $n \in \mathbb{Z}$. If n is even, then n^2 is divisible by 8.

Section D — Writing

6. In five to eight complete sentences, explain why an element that fails the hypothesis of an implication cannot serve as a counterexample. Use the truth table for $\Rightarrow$ and one specific example.

Section E — Challenge (optional)

7. Disprove: for every positive integer n, the number $n^2 + n + 41$ is prime.